Lecture Notes on

Geometry and Topology

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Notes on geometry and topology. Resources used:

- Frederic Schuller's *Geometrical Anatomy of Theoretical Physics* video lectures.
- Bott and Tu's *Differential Forms in Algebraic Topology.*
- Littlejohn's *Physics 250* lecture notes

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1 Geometry

1.1 Manifolds

1.2 Differential constructions

Define Ω^* to be the $\mathbb{R}$-algebra generated by the symbols $dx^1, \ldots, dx^n$ satisfying $dx^i dx^i = 0$ and $dx^i dx^j = -dx^j dx^i$[1]. We define the **vector space of C^∞ differential forms** to be

$$\Omega^*(M) = C^\infty(M) \otimes_{\mathbb{R}} \Omega^*.$$

A **differential k-form** is then a function in $\Omega^*(M)$ that has k dx^i's in it, i.e. $\omega = \omega_I dx^I$ for some multi-index $I = (i_1, \ldots, i_k)$. We can rewrite $\Omega^*(M)$ in terms of these spaces as $\Omega^*(M) = \bigoplus_{k=0}^{\dim M} \Omega^k(M)$; this is just saying that you can partition this space into subspaces based on the degree of the differential form. Note the upper summand: by anti-symmetry, $k \leq \dim M$ (because otherwise you'd have $dx^i \wedge dx^i$ terms in your form).

> **Remark 1.1.** There is a clear similarity here between $\Omega^*(M)$ and the Fock space $\mathcal{F} = \bigoplus_{n=0}^{\infty} \mathcal{H}_n$ by analogizing n-forms to n-particle states. The similarity increases further when you consider a fermionic Fock space, where you directly identify fermionic fields $\psi \leftrightarrow dx$.

There are two natural constructions on the space of differential forms: the **wedge product** and the **exterior derivative**. The wedge product is just the natural way of multiplying forms; it's directly given by

$$\omega \wedge \eta = \omega_I \eta_J dx^I dx^J.$$

By definition, we can see that $\omega \wedge \eta = (-1)^{\deg \omega \deg \eta} \eta \wedge \omega$. The natural derivation on $\Omega^*(M)$ is the **exterior derivative**, $d \colon \Omega^n(M) \to \Omega^{n+1}(M)$, defined by

- If $f \in \Omega^0(M)$, then $df = \partial_i f dx^i$.
- If $\omega \in \Omega^k(M)$, then $d\omega = df_I dx^I$.

The two most important properties of the exterior derivative are that it's nilpotent and that it obeys graded Leibniz. That is,

$$d^2 \equiv 0 \qquad\qquad d(\omega \wedge \eta) = d\omega \wedge \eta + (-1)^{\deg \omega} \omega \wedge d\eta.$$

Notice that the minus sign in the second term only depends on the degree of the first form in the product. This is because you have to anti-commute the d only past ω, thus moving it past $\deg \omega$ dx's.

> **Example 1.1** (Div, grad, curl, and all that)
>
> With the exterior derivative we can regain the definitions of div, grad, and curl extremely easily. On $\mathbb{R}^3$, $\Omega^0(M) \cong C^\infty(\mathbb{R}^3) \cong \Omega^3(M)$, and $\Omega^1(M) \cong \Omega^2(M)$, so we can identify these spaces with one another. Let $f \in \Omega^0(M)$. On functions,
>
> $$df = \partial_x f \, dx + \partial_y f \, dy + \partial_z f \, dz \leftrightarrow \nabla f$$

[1] This second condition is actually necessary, as we're dealing with the multiplication of the symbols themselves; we can't just apply condition (1) to $dx^i + dx^j$ and get condition (2) for free.

On one-forms,

$$d(f_i\,dx^i) = (\partial_x f_y - \partial_y f_x)\,dx \wedge dy + (\partial_y f_z - \partial_z f_y)\,dy \wedge dz + (\partial_z f_x - \partial_x f_z)\,dz \wedge dx \leftrightarrow \nabla \times f$$

And on two-forms,

$$d(2\partial_{[i} f_{j]}\,dx^i \wedge dx^j) = (\partial_x f_x + \partial_y f_y + \partial_z f_z)\,dx \wedge dy \wedge dz \leftrightarrow \nabla \cdot f$$

In summary,

$$\begin{aligned} d(\text{0-forms}) &= \text{gradient}, \\ d(\text{1-forms}) &= \text{curl}, \\ d(\text{2-forms}) &= \text{divergence}. \end{aligned}$$

Pretty cool!

(talk about cohomology theory here, then rewrite the first part of this talking about how getting to cohomology is one of our explicit goals)